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AN IGNITION TRANSIENT MODEL FOR SOLID PROPELLANT ROCKET MOTORS

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Ignition transients are a very important practical aspect in the design of solid propellant rockets, for the prediction and control of dynamic loads on the motor and on the entire vehicle. The present study is detailing a simple model and the relative numerical solution method for the analysis of ignition transient in solid propellant motors. The model is capable of providing an accurate and reliable prediction of the most significant engineering quantities (pressure and velocity in the thrust chamber) for different rocket configurations, for which currently available methods are yielding relatively unsatisfactory results: this would allow for an effective preliminary design of the motor, minimizing the need for experimental tests. A theoretical quasi-one-dimensional model is derived from basic conservation laws, taking into account the initiation of combustion by the igniter discharge signal, the heat transfer from igniter combustion products to the propellant grain surface, the flame spreading flow field in the chamber and the increase of chamber pressure up to to the quasi-steady operating condition. The set of modeling equations are then solved using a slightly modified Lax-Friedrichs method. Sample results are presented for typical solid boosters of the Ariane family (Ariane 4 and Ariane 5).

List of Symbols

A	combustion chamber cross section
a	pre-exponential factor in the non erosive burning rate
α	thermal diffusivity
β	erosive burning exponent
c_p	constant pressure specific heat
C_{hc}	experimental factor for correction of convective heat-transfer coefficient
γ	ratio of specific heats
D_h	hydraulic diameter
e	specific energy
h	specific enthalpy of combustion gases
h_c	local heat transfer coefficient
k	thermal conductivity
k_{eb}	erosive burning constant
n	pressure exponent in the non erosive burning rate

μ	dynamic viscosity
P	combustion chamber perimeter
p	pressure
R	universal gas constant
r_b	burning rate
ρ	gas density
ρ_{pr}	grain density
σ_{pr}	solid propellant thermal sensibility
T	temperature
τ_{xx}	normal shear stress
τ_w	shear stress on the wall
u	gas velocity
v_{inj}	igniter gas velocity
x	axial coordinate
y	radial coordinate into the propellant grain (depth)

Subscripts

b	burning condition
brg	burning rate reference condition
f	adiabatic flame condition
i	ignition value
ig	igniter
p	port

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p_g	propellant gas phase
p_i	propellant ignition value
p_r	propellant condensed phase
p_s	propellant surface
p_{ref}	reference condition
X_E	nozzle entrance
$stap$	nozzle uncapping
t	nozzle throat
w	wetted
0	total

Introduction

The ignition transient in Solid propellant Rocket Motors (SRM's) is usually defined as the time interval between the application of the ignition signal and the instant at which the rocket motor attains its equilibrium or design operating conditions. The ignition transient can be generally considered as composed by three phases: 1) the induction period for first local ignition, 2) the period for flame spreading over the exposed surface of the propellant grain, and 3) the chamber filling period. The first ignition and flame spreading periods are preceded by the heating of the propellant grain by igniter products.

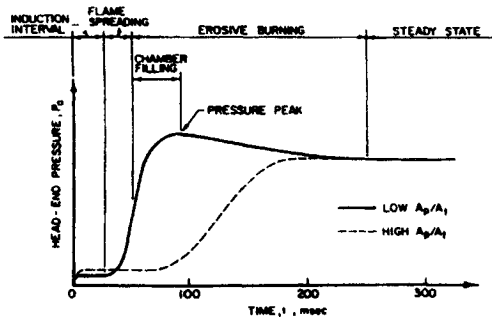


Fig. 1 Typical transient of SRM

More in detail, the ignition transient comprises a complex series of correlated events/processes such as 1) the initiation of the igniter discharge signal, 2) the heat generation caused by chemical reactions between igniter species, 3) the conductive, convective, and radiative heat transfer from the igniter products to the propellant surface, 4) the flame spreading flow field in the chamber, and 6) the increase of chamber pressure up to the quasi-

steady operating condition. These events are sometimes accompanied by different undesired phenomena, such as overpressures, hang-fires (delayed ignition), damaging shock waves (detonation), combustion oscillations, chuffing and combustion extinction.

Many studies, both experimental and theoretical, have been performed on ignition transient aspects in SRM's, which have been helpful in understanding and interpreting many fundamental processes and in the successful design of rocket motors. However, a few aspects of flame spreading process and of the ignition transient are still poorly understood and described. Even though many experimental activities have assisted in providing considerable physical insight into this complex problem, the major aim of the research has been toward the development of predictive models in order to reduce the cost associated with experimental "trial and error" design techniques. Moreover, because of the multitude of different designs for new rocket motors, extrapolation from previous empirical data is often unreliable.

For ignition transient, available models consider both temporal and spatial variations. As the igniter mass flow rate is considered as a known input parameter in all existing models, it is essential to accurately measure and quantify it. Erosive burning effects must also be included for an accurate prediction of ignition transient, whilst dynamic burning may or may not be important, depending upon combustion conditions: it is generally believed that dynamic burning effects are relevant only if the thermal wave penetration depth in the propellant grain is larger than that corresponding to the steady-state burning conditions.

Ignition transient models in which properties are assumed to be spatially uniform [$p(t)$ models]¹ can be used for high port-to-throat area ratio motors. However these models are recommended for trend analysis only, and not for detailed design calculations. On the other hand, constant time models [$p(x)$ models] are inadequate and should not be used. Multidimensional models^{2,3} are more accurate and the results are usually in good agreement with the available experimental data, but at the cost of long computation time and pre/post processing.

Finally, transient models which consider only one spatial coordinate^{4,5,6,7,8} are both computationally inexpensive and adequate to describe the ignition transient of rocket motors with high volumetric loading density, small port-to-throat area ratio and large length-to-diameter. However, for such motors the chamber geometric parameters cannot be considered as uniform throughout the motor, nor can they be treated in a quasi-steady way if the entire ignition transient is to be predicted. This latter type of model was selected for the present study and is detailed in the following paragraphs.

The Physical Model

The basic assumptions used in the model are as follows: 1) all chemical reactions occur on the propellant surface in a thin planar zone; 2) flow in the chamber is one-dimensional; 3) rate processes at the propellant surface are quasi-steady, in the sense that their characteristic times are short compared to the characteristic time of the pressure transient; 4) the chamber gas obeys the perfect gas law.

It should also be noted that the solid phase is modeled as a non reactive material until ignition: local ignition is assumed to occur at the attainment of a critical temperature at the propellant surface, and flame spreading is defined by the sequential ignition of propellant surface elements.

The mathematical formulation of the model consists of the following equations: 1) mass, momentum and energy conservation equations in unsteady, one-dimensional form for the gas phase with unstable source terms; 2) equations of state for the perfect gas flowing in the motor; 3) proper initial conditions at the beginning of the transient; 4) boundary conditions at the fore-end and at the aft-end of the motor combustion chamber; 5) a semi-empirical correlation for the convective heat transfer due to the highly turbulent flow in the port; 6) a burning rate law for the solid propellant grain, including the effects of initial temperature, pressure and velocity (erosive burning); 7) an unsteady solid phase temperature equation for determining propellant surface temperature during the induction interval, coupled with an appropriate ignition criterion for the solid propellant.

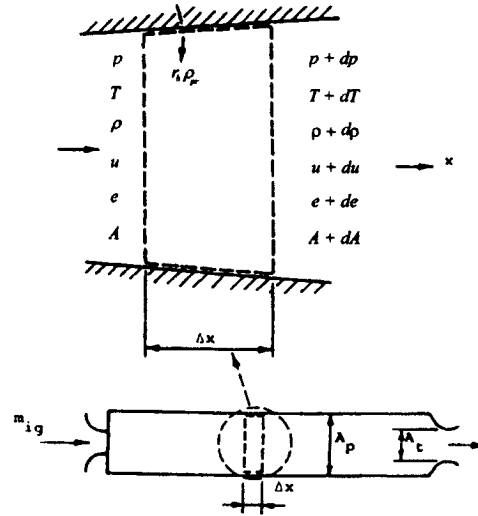


Fig. 2 Control volume definition

The governing gas phase equations under the above assumptions are:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\rho u}{A_p} \frac{\partial A_p}{\partial x} = \frac{r_p P_b}{A_p} \rho_p$$

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\rho u^2}{A_p} \frac{\partial A_p}{\partial x} = -\frac{\partial p}{\partial x} - \frac{r_p P_b}{A_p} \rho u$$

$$\frac{\partial(\rho e_T)}{\partial t} + \frac{\partial(\rho u e_T)}{\partial x} = -\frac{\partial(pu)}{\partial x} + \frac{r_p P_b}{A_p} \rho_p h_f$$

These conservation equations do not include a few terms, which were found to be negligible by an order of magnitude analysis. In particular the following were neglected: the rate of (convective) heat loss to propellant and non-propellant port walls; the rate of mass accumulation in the free volume created by the propellant surface regression (as compared to the rate of mass addition due to burning); the viscous stresses; the axial heat conduction. The initial conditions for the gas phase equations are:

$$\rho(x,0) = \rho_i; \quad u(x,0) = u_i; \quad e_T(x,0) = e_{T,i}$$

The above set of governing equations constitutes a sixth-order hyperbolic system, requiring six boundary conditions for the solution. The latter can be derived from the compatibility relations at the boundaries, which in turn are obtained by solving the governing equations by the method of characteristics.

The compatibility relation may be found in any good book on computational techniques⁹, and will not be repeated here. A sample of the numerical calculation grid and characteristic direction at the downstream boundary is shown in Fig. 3.

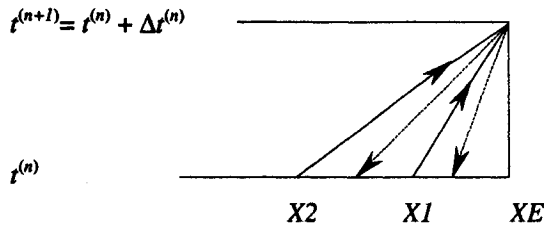


Fig. 3 Characteristic time scheme

Propellant surface temperature

The propellant surface temperature¹⁰ before the ignition onset is obtained by solving the 1-D unsteady heat equation for the solid phase, normally to the grain surface using the BI method. The resulting ordinary differential equation, which describes the variation of propellant surface temperature with time at any axial location, is given by:

$$\frac{dT_{ps}}{dt} = \frac{4\alpha_{pr}h_c^2(T - T_{ps})^3}{3k_{pr}^2(T_{ps} - T_{pi})(2T - T_{ps} - T_{pi})}$$

The singularity which occurs at $t=0$ in this equation is avoided by remapping the time coordinate with a small origin shift: $T_{ps}(0) = T_{pi} + \epsilon_T$.

Empirical heat transfer correlation

The local convective heat transfer coefficient h_c is deduced from the conventional Dittus-Boelter correlation for turbulent flow in pipes:

$$h_c = \left[0.026(\rho u)^{0.8} \left(\frac{\mu}{D_H} \right)^{0.2} \text{Pr}^{0.6} c_p \right] C_{hc}$$

where C_{hc} is an experimental factor accounting for the contribution of radiative and conductive heat transfer, the effects of geometry and jet-impingement^{11,12,13,14,15}.

In order to properly define the C_h factor for the present study, experimental data from two Ariane 4 motors tests have been considered, and the appropriate correction factor computed. The results are shown in , which highlights the good

repeatability for the factor.

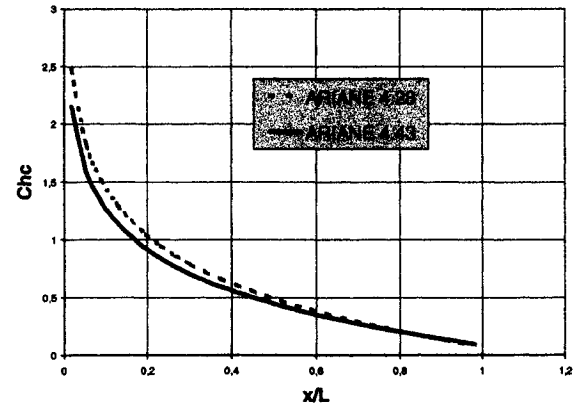


Fig. 4 Experimental correction factor C_{hc}

The differences, even though both motors are nominally equal, are a consequence of the different temperature and humidity conditions at the time of grain firing.

Burning rate law

A modified Lenoir-Robillard's burning rate law^{16,17} is used to account for erosive burning:

$$r_b = ap^n + k_{eb} h_c e^{\frac{\beta r_b \rho_{pr}}{\rho u}}$$

where the erosive burning constant k_{eb} is estimated from theoretical considerations¹¹, and the erosive burning exponent β is evaluated from water-quench experiments and adjusted to fit the experimental pressure time curves for all values of A_p/A_t tested.

The Solution Algorithm

An appropriate computer program was implemented in order to obtain numerical solutions of the described model, by the simultaneous integration of the three governing equations and the propellant surface temperature equation. Since the set of governing equation is hyperbolic in time, the numerical solution is based on a finite volume discretization in space (x) and generalized explicit method in time (t).

The vector form of the implemented equations is:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{S}$$

where:

$$\mathbf{U} = [\rho, \rho u, \rho e_T]^T$$

$$\mathbf{F} = [\rho u, (\rho u^2 + p), u(\rho e_t + p)]^T$$

$$\mathbf{S} = \left[\frac{r_b P_b}{A_p} \rho_{pr}, \left(\frac{p}{A_p} \frac{\partial A_p}{\partial x} - \frac{r_b P_b}{A_p} \rho u \right), \frac{r_b P_b}{A_p} \rho_{pr} h_f \right]^T$$

The equations are solved by a Lax-Friedrics method^{9,18}, modified to satisfy the stability, consistency and convergence conditions on a variable step grid.

Ariane 4 SRM Results

The first test case for the code was the simulation of a static firing of the Ariane 4 SRM.

The experimentally measured igniter mass flow rate used in this calculation is shown in Fig. 6.

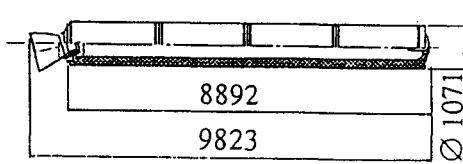


Fig. 5 Ariane 4 SRM

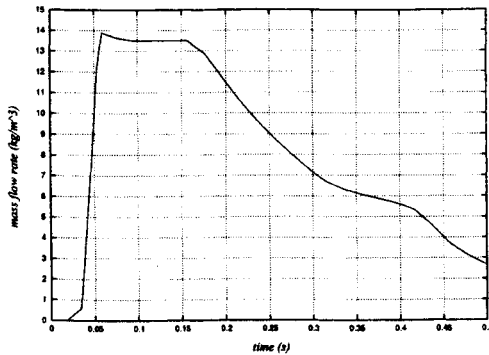


Fig. 6 Ariane 4 igniter mass flow rate

The relevant results are shown in Figs. 7-11, including the simulated pressure $p(x,t)$, temperature $T(x,t)$, density $\rho(x,t)$, and velocity $u(x,t)$ of the combustion gas in motor chamber, and the propellant surface temperature $T_{ps}(x,t)$ which is representative of the flame spreading process.

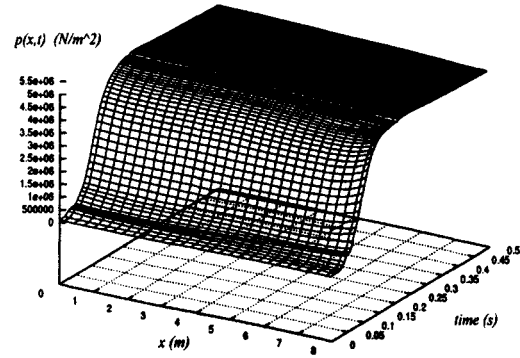


Fig. 7 Ariane 4 chamber pressure $p(x,t)$

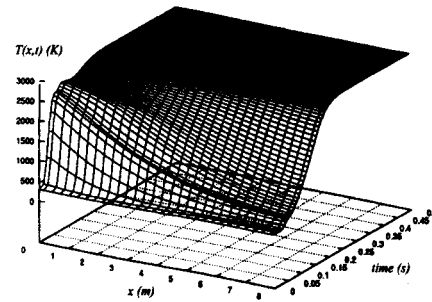


Fig. 8 Ariane 4 gas temperature $T(x,t)$

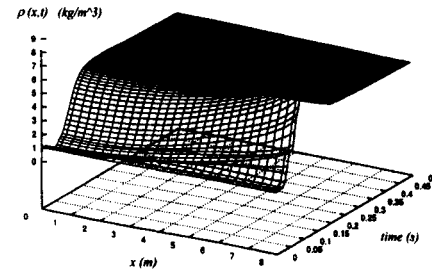


Fig. 9 Ariane 4 gas density $\rho(x,t)$

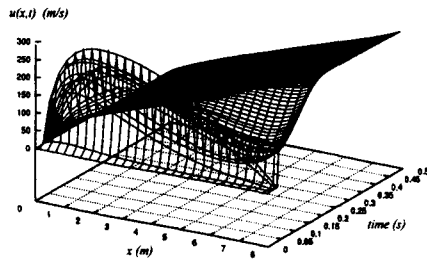


Fig. 10 Ariane 4 gas velocity $u(x,t)$

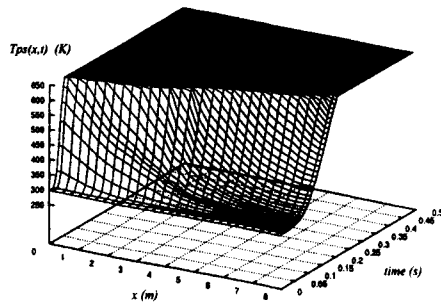


Fig. 11 Ariane 4 propellant surface temperature

$$T_{PS}(x,t)$$

The simulated chamber pressure $p(0,t)$ at the fore-end section and the node ignition sequence are shown together in Fig. 12.

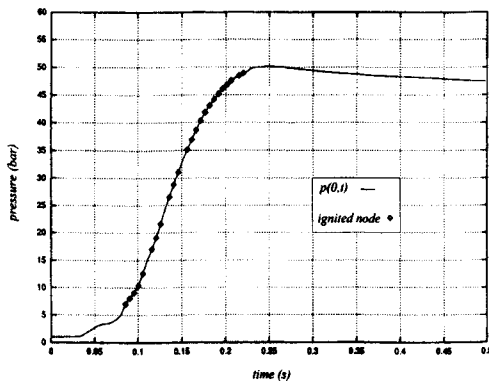


Fig. 12 Ariane 4 chamber pressure $p(0,t)$

For comparison and validation purposes, Figs.

13-14 are showing the difference between the experimental data obtained from an Ariane 4 SRM and the simulated chamber pressure $p(0,t)$, implementing combustion models respectively not including and including erosive burning effects.

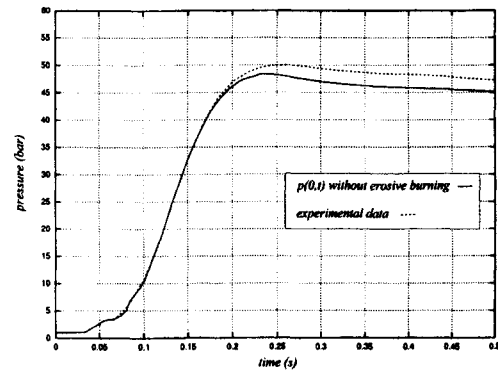


Fig. 13 Ariane 4 chamber pressure $p(0,t)$ without erosive burning and experimental data

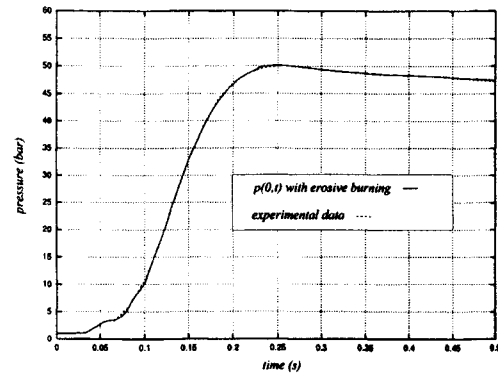


Fig. 14 Ariane 4 chamber pressure $p(0,t)$ with erosive burning and experimental data

Ariane 5 SRM Results

The second test case for the model and the code implementing it was a static firing of the Ariane 5 SRM. Because of the geometric and scale difference with the Ariane 4 cases which were used in order to calibrate the different model parameters, the Ariane 5 SRM case can be considered as a more difficult validation case, properly testing the model applicability to a wide range of different motors. The most interesting results are presented in Figs. 15-16.

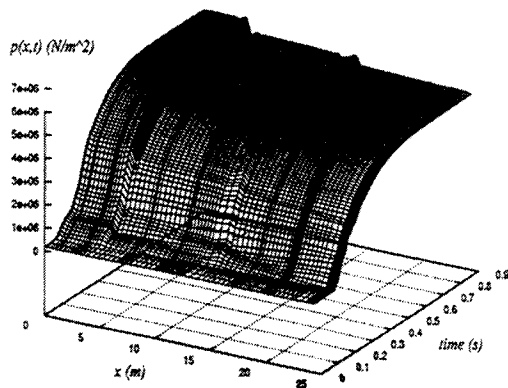


Fig. 15 Ariane 5 chamber pressure $p(x,t)$

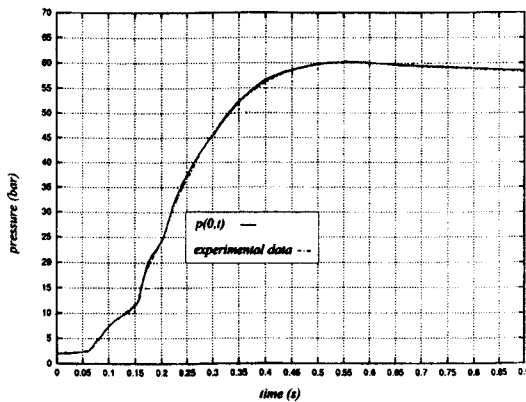


Fig. 16 Ariane 5 chamber pressure $p(0,t)$ with erosive burning and experimental data

The effect of the segmented grain configuration in the Ariane 5 SRM is clearly visible in the above figures, as a discontinuity of pressure corresponding to the grain slots.

The Quasi-Steady Burning Phase

The above model is more generally applicable to the whole burning phase of a solid rocket motor, if a few additional considerations are introduced.

In particular, two modifications of reference burning rate should be accounted for: the bulk burning rate scale factor and the variation of burning rate within the cured propellant grain. The latter can be caused by processing and/or migration of liquid species to propellant interfaces.

Bulk burn rate scale factors exist between propellant samples (tested in subscale motors) obtained from subscale mixers and propellant grains

obtained in full scale mixers: these scale factors have been observed to be as large as 3 or 4%. Furthermore, the scale factor between subscale and full scale motors has been observed to be as large as 66%. This scale factor uncertainty represents a potential amplitude error in the predicted pressure level, but is not time or burn distance dependent.

On the other hand, the burn rate variation within the cured propellant grain represents a more complex problem, since it is not easily predictable and creates distortions in the surface area vs. burn distance relationship predicted on the basis of simple uniform burn-back. The mass generation rate, and hence the pressure-time curve, will deviate from classical predictions because of both the burn rate and surface area deviations. This phenomenon has been referred to by various terms such as: "hump effect", BRAF^{19,20,21} (Burning Rate Anomaly Factor), and "residuals effect". In general, all the errors which are reflected in curve shape deviations are lumped into a factor, which is defined as the ratio of the actual to the predicted pressure taken to the 1-n power, to relate it to an effective surface area or effective burn rate at a given time or burned web distance.

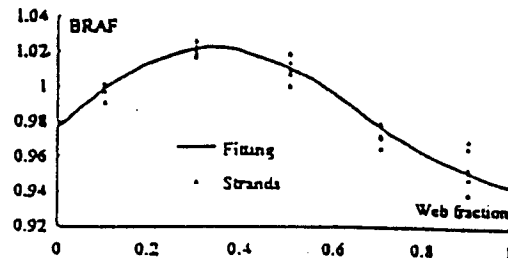


Fig. 17 Burning Rate Anomaly Factor

The burning life of an Ariane 4 SRM (up to burn-out) was computed using the model described, both including a BRAF correction taken from the literature (Fig. 17) and not including it. The results are shown in Figs. 18-19.

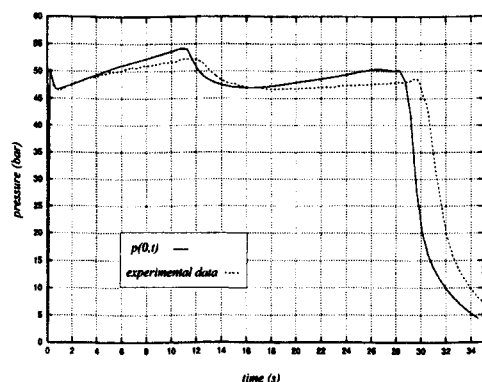


Fig. 18 Ariane 4 chamber pressure $p(0,t)$ without BRAF and experimental data

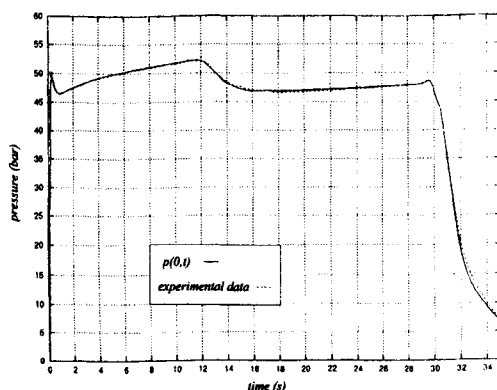


Fig. 19 Ariane 4 chamber pressure $p(0,t)$ with BRAF and experimental data

Conclusions

As shown by the very good accuracy obtained in both the Ariane 4 and 5 cases, the simple quasi-one-dimensional model herein described and its numerical implementation are capable of providing an accurate and reliable prediction of the ignition transient of Solid propellant Rocket Motors. The code is very inexpensive from the computational point of view and may therefore be used as an effective design tool since the very early phases of motor conception.

Moreover, if the necessary experimental data are available, the model may also be used for the prediction of the motor performance during the different phases of operation: ignition transient, quasi-steady state and tail-off.

Some of the main characteristics (and positive aspects) of the model are:

- igniter propellant may be different from main motor propellant;
- jet-impingement effects are taken into account;
- nozzle throat section erosion is included;
- propellant grain may be monolithic or segmented.

The ballistic data used in the code, which is based on the analysis of tests performed on different motor classes, have been validated for $p(x)$ -type models. The proper application to $p(x,t)$ -type models, spanning the whole burn life of the motor, will need a careful assessment and validation process, using new accurate experimental data, in order to account for the mass-budget, the thrust-imbalance and extinction phase.

However, even using the available data, differences between numerical results and experiments are less than 1% for both the Ariane 4 and Ariane 5 cases, indicating that the model is likely to be performing very well for many different motor types. This is confirming the fact that a correct physical model leads to good results even when simple numerical methods are used

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